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Beyond Volume: A Legitimacy-Theory Analysis of the Substance of Corporate Sustainability Reporting

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ABSTRACT

This study examines whether the expansion of corporate sustainability reporting reflects substantive improvement or legitimacy-oriented disclosure. Using hand-collected sustainability reports from 364 U.S.-listed Forbes Global 2000 firms (2018–2023), we classify GRI indicators by news direction and sensitivity and estimate firm fixed-effects models across a pre-COVID period (2018–2019), an acute COVID period (2020–2021), and a post-acute phase (2022–2023). While reporting volumes increased—especially for social indicators—coverage remains far below GRI benchmarks, particularly in sensitive areas. High-impact industries exhibit stronger growth in environmentally sensitive disclosures during the pre-COVID and acute COVID periods, but growth decelerates in the post-acute phase and between-industry differences largely diminish. Overall, disclosure expansion is disproportionately concentrated in positive and non-sensitive indicators, consistent with legitimacy-driven reporting rather than substantive accountability.

1 | Introduction

Public attention to sustainability has intensified as climate risks and social inequities escalate (Ranjbari et al. 2021). Climate change is now recognised as a leading global risk, with potentially irreversible consequences for growth and poverty (Deutsche 2020; Tol 2018). Policy momentum has likewise accelerated through the United Nations' 2030 Agenda and Sustainable Development Goals and the Paris Agreement's commitment to limit warming well below 2°C (van der Waal and Thijssens 2020).

These developments heighten legitimacy pressures on firms to address societal and environmental impacts and to communicate those responses through corporate sustainability reporting (SR) (Siddiqui and Uddin 2016). One response has been the rapid expansion of SR to communicate economic, social and environmental activities and to shape reputation and legitimacy (Higgins and Coffey 2016). Reports are also positioned as vehicles for demonstrating contributions to the SDGs and

Paris climate agreement goals and for navigating contentious issues during crises (Bellucci et al. 2021; Zharfpeykan and Akroyd 2022). Globally, reporting rates surged through the 1990s and early 2000s but have since slowed; since 2020, coverage has largely plateaued (79% of N100; 96% of G250), with GRI remaining the dominant standard (KPMG 2024).

The COVID-19 pandemic constituted a multifaceted shock across sustainability's economic, social and environmental pillars (Boiral et al. 2021) and intensified stakeholder expectations for demonstrable commitment (Zharfpeykan and Ng 2021). Unlike climate change as a long-horizon risk, COVID-19 was a sudden disruption that plausibly reweighted disclosure priorities toward vulnerable stakeholders and near-term resilience (Parker and Narayanan 2023). Prior research shows that under crisis firms adjust disclosure type, quality and tone to manage legitimacy (Cho et al. 2010; O'Donovan 2002; Maroun 2018), yet COVID-era evidence is mixed, pointing to shifts toward short-term economic priorities (Klymenko and Lillebrygfjeld Halse 2022), increased social attention (Parker and Narayanan 2023)

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and possible environmental repositioning (Markard and Rosenbloom 2020). The direction of reporting change was therefore not predetermined.

Although KPMG (2024) documents a plateau in reporting incidence beyond the acute phase of the pandemic, it does not assess whether disclosure content and quality shifted. This is a critical gap, as stable volume does not imply stable substance. Volume alone cannot reveal whether firms are engaging with material impacts or sidestepping sensitive areas. Assessing disclosure composition—what indicators are prioritised, expanded or omitted—offers a more accurate test of whether practices are evolving toward substantive transparency or remain instruments of legitimacy management (Cho et al. 2010; O'Donovan 2002).

We therefore shift attention from how much firms report to what they report. Using GRI-based indicators, we examine whether expanded reporting reflects broader, more candid engagement with material topics, especially negative and sensitive categories or whether it primarily amplifies 'good-news' and neutral items. In this regard, we treat the COVID-19 pandemic as an exogenous shock that serves as a stress test of the maturity and resilience of corporate SR strategies before, during and after the acute phase of the pandemic, while recognizing that pandemic-related effects may persist beyond the immediate crisis period.

To identify reporting strategies, this study hand-collected and analysed corporate sustainability reports from 364 U.S.-listed Forbes 2000 companies that used Global Reporting Initiative (GRI) standards between 2018 and 2023. Building on Zharfpeykan (2021), we refine the classification of indicators through a legitimacy-theory lens (good, bad, neutral; with flags for sensitive stakeholder-salient areas). We estimate firm fixed-effects models to identify within-firm changes over time, compare environmentally high-impact versus low-impact industries, and distinguish pre-COVID (2018–2019), acute COVID (2020–2021), and post-acute (2022–2023) phases.

This study contributes to our understanding of corporate SR strategies in some key areas. First, by moving from quantity to composition, this study provides granular evidence on which indicators firms disclose and how that mix shifts, addressing the risk that rising volumes create an illusion of progress (KPMG 2024) without fuller engagement with sensitive topics (Cho et al. 2010). Second, by coding indicator news and sensitivity, we show how legitimacy is managed through selective composition, and how responses differ by industry exposure (Siddiqui and Uddin 2016; Zharfpeykan and Akroyd 2022). Third, this research contributes to the crisis literature by investigating how corporate SR strategies evolved during and after the acute phase of the COVID-19 pandemic. According to crisis communication theory, a crisis is 'an unpredictable event that threatens important expectations of stakeholders and can seriously impact an organisation's performance, leading to negative outcomes' (Coombs 2007, 2). Crises can vary by type and the degree of reputational threat they pose (Coombs 2007). While previous studies have explored the impact of the 2008 Global Financial Crisis on corporate SR, with mixed findings (e.g., Miras Rodríguez et al. 2014), other research has focused on how firms adjust their reports following different scandals (e.g., Allam et al. 2020; Boiral et al. 2022). COVID-19, however,

represents a 'victim cluster' crisis, where firms have limited responsibility for the cause and face only mild reputational risks (Zharfpeykan and Ng 2021). Treating COVID-19 as an exogenous shock (Coombs 2007; Zharfpeykan and Ng 2021), we examine whether crisis-driven adjustments reflect transient, symbolic responses or more durable, substantive shifts (Parker and Narayanan 2023; Markard and Rosenbloom 2020). Last, this study provides practical insights for managers and stakeholders regarding how firms can adjust their SR strategies to provide more substantive reporting that better addresses external pressures. For managers, it offers guidance on aligning reporting practices with global sustainability expectations and enhancing credibility. For stakeholders, it improves their ability to assess the transparency and relevance of corporate disclosures.

The rest of this paper is structured as follows. The next section reviews the prior literature. Section 3 explains the research method and content analysis procedures, while Sections 4 and 5 present and discuss the results.

2 | Literature Review

2.1 | Corporate Sustainability Reporting Strategies

Sustainability has moved to the centre of corporate strategy, shaping how firms set objectives, allocate resources and communicate progress (Beyne et al. 2021). In this context, corporate SR refers to how firms share information about sustainability initiatives with internal and external stakeholders (Gallardo-Vázquez et al. 2025). The literature recognises SR's strategic value but also documents wide variation in the extent and quality of disclosure (Troshani and Rowbottom 2024). Firms pursue different reporting strategies, from trust-building and community engagement to signalling environmental stewardship and risk management, often reflecting attempts to manage stakeholder expectations and preserve organisational legitimacy (Borgstedt et al. 2019).

Legitimacy theory provides a foundational lens to interpret these choices. Organisations seek congruence between their activities and evolving societal norms; disclosure becomes a means to maintain, repair or extend legitimacy (Deegan 2019). Ashforth and Gibbs (1990) distinguish symbolic responses, gestures and narratives that signal alignment, from substantive responses that change underlying practices. In corporate SR, symbolic responses include greenwashing, where firms accentuate positive outcomes and downplay negatives to construct an impression of responsibility (Hsiao et al. 2022; Melloni et al. 2017). Greenwashing research further clarifies this behaviour as the strategic use of positive communication that is not matched by underlying performance. Delmas and Burbano (2011) define greenwashing as the combination of poor environmental performance and positive environmental communication, highlighting how firms selectively emphasise favourable information. Lyon and Montgomery (2015) conceptualise greenwashing more broadly as communication that leads stakeholders to form overly positive impressions, often through selective disclosure and emphasis on positive claims. Together, this literature suggests that an emphasis on good and neutral disclosures, alongside limited engagement with negative or sensitive issues, is a

core manifestation of greenwashing. By contrast, substantive reporting acknowledges trade-offs, discloses negative outcomes and documents remediation.

The distinction between symbolic and substantive responses has implications for how scholars assess reporting quality. O'Donovan (2002) observed an evolution from purely symbolic statements toward more substantive disclosures as legitimacy pressures intensified. Yet, more recent work shows that firms can still avoid sensitive topics or present selective narratives (Zharfpeykan 2021). Counting pages or indicators can therefore confuse more disclosure with better disclosure.

This concern is consistent with Clarkson et al. (2008), who developed a GRI-based content index to distinguish between 'hard' (performance-based, verifiable) and 'soft' (symbolic, narrative) environmental disclosures. They show that socio-political theories are more effective at explaining what firms say than how much they say, highlighting the importance of examining disclosure composition rather than volume alone. Their findings motivate the need for indicator-level approaches that move beyond counts toward the nature and credibility of disclosed information.

A composition-based view—examining indicator types, their news direction (good, bad, neutral) and sensitivity—more directly reveals whether reporting reflects genuine accountability or curated legitimacy management. Thus, corporate SR functions simultaneously as an accountability mechanism and a tool of perception management (Beck et al. 2010), and the academic challenge is to empirically distinguish these roles.

Rather than asking whether firms disclose more, this study examines whether recent expansions in SR reflect a shift toward substantive accountability or primarily symbolic legitimacy management. By analysing indicator-level disclosure composition across environmental exposure groups and crisis phases, we assess whether growth is accompanied by broader engagement with sensitive and adverse indicators or instead is concentrated in favourable and low-risk categories. Empirically, environmentally sensitive industries tend to face stronger scrutiny and thus exhibit more extensive disclosure, including on difficult topics (Rodrigues and Mendes 2018). If SR is becoming more substantive rather than merely expansive, this pressure should be reflected not only in higher disclosure volume for high-impact firms but in broader engagement with environmentally sensitive and adverse indicators. Accordingly, this study hypothesises that:

H1. *Environmentally high-impact firms exhibit stronger growth in environmental bad-news and sensitive indicators than low-impact firms.*

2.2 | Corporate Sustainability Reporting, Legitimacy Management and Crisis

Firms operate amid recurrent shocks—natural disasters, financial crises, energy volatility, supply-chain disruptions—that alter stakeholder expectations and risk profiles (MacKenzie et al. 2012; Zdolšek and Beloglavec 2023). Under such

conditions, corporate SR becomes a vehicle to reassure stakeholders, demonstrate responsiveness and protect reputation (Zharfpeykan and Ng 2021). Sustainability reports have been used to guide stakeholders' interpretations during disputes and controversies (Bellucci et al. 2021), to articulate strategic priorities and targets, and to enable deeper engagement with diverse audiences (Zdolšek and Beloglavec 2023).

Legitimacy theory anticipates that crises heighten pressures for disclosure but do not predetermine whether that disclosure will be symbolic or substantive. Organisations can respond symbolically (e.g., positive tone, selective metrics) or substantively (balanced coverage, including 'bad-news' items). Prior research shows that firms adjust disclosure type, quality and tone to manage legitimacy during controversy (Cho et al. 2010; O'Donovan 2002; Maroun 2018). In some contexts, firms may avoid potentially damaging information to limit liability and protect their image (Jin and Austin 2022). Philippe and Durand (2011), Vergne (2012) and Yang et al. (2020) describe how organisations deploy proactive and reactive mechanisms—framing, categorisation, buffering—to mitigate negative evaluations. From a measurement standpoint, these dynamics imply that volume metrics can rise sharply during crises while disclosure composition remains strategically selective.

Global diffusion of corporate SR has been well documented. KPMG (2024) reports strong growth in reporting through the 1990s and early 2000s, followed by slower increases and signs of plateau since 2020, and GRI remains the dominant framework. Yet such incidence statistics leave a central question: did the expansion come with more substantive reporting? Zharfpeykan and Akroyd (2023) find that disclosure quality on material issues declined from 2011 to 2019 among Australian mining, metals and financial services firms, which increasingly adopted greenwashing strategies. Székely and vom Brocke (2017) found that the most frequently mentioned topics in global sustainability were emissions and resource consumption under the environmental dimension, employees under the social dimension and financial data under the economic sustainability dimension. However, they did not examine shifts in the indicator mix or the news/sensitivity of disclosure.

Taken together, this literature suggests that crises create strong incentives for firms to expand sustainability communication, but provide no assurance that such expansion entails greater engagement with sensitive or adverse issues. This opens an empirical question not only about whether firms disclosed more during COVID-19, but also about how they reweighted the composition of disclosure across sustainability domains and indicator types.

2.3 | Corporate Sustainability Reporting and the COVID-19 Pandemic

On 11 March 2020, the World Health Organisation declared COVID-19 a global pandemic, precipitating health, social and economic dislocation (Carnegie et al. 2023). The pandemic affected all three pillars of corporate sustainability (Boiral et al. 2021), stressing employees and supply chains (Ikram et al. 2020) and exposing interdependencies among public health, poverty, climate and financial stability (Adams and Abhayawansa 2022).

Stakeholder expectations evolved rapidly: firms were expected to protect health and safety, preserve employment where possible and maintain essential services (He and Harris 2020). Investors intensified their attention to ESG performance (Pástor and Vorsatz 2020), suggesting a potential increase in corporate SR salience.

Theoretically, COVID-19 represents a victim-cluster crisis in which firms had limited responsibility for the precipitating event and thus faced milder attribution of blame (Zharfpeykan and Ng 2021; Coombs 2007). This framing shapes disclosure incentives by intensifying pressures to communicate responsiveness without necessarily compelling substantive accountability. Some studies suggest a short-term pivot toward economic survival (Klymenko and Lillebrygfjeld Halse 2022), while others document expanded attention to social topics—health, safety, labour practices—during the pandemic (Parker and Narayanan 2023). Markard and Rosenbloom (2020) argue that a crisis can also catalyse low-carbon transitions. Empirically, Phang et al. (2023) find that higher sustainability disclosure during COVID-19 was associated with higher market valuations in Australia, particularly among loss-making firms, consistent with SR's insurance-like reputation effects. Yet many COVID-era studies rely on extent or textual frequency measures, without interrogating whether pandemic-era increases extended to sensitive or adverse indicators.

From a legitimacy perspective, crises heighten incentives to communicate but do not guarantee a shift toward substantive transparency. Prior studies document heightened attention to social and employee-related issues during COVID-19 (Parker and Narayanan 2023). This suggests a likely reweighting of disclosure priorities toward social domains, without necessarily a proportional expansion into sensitive and adverse indicators. Accordingly, consistent with a legitimacy-management interpretation, we hypothesise that:

H2. *The COVID-19 period is associated with an increase in social sustainability disclosure, without a commensurate expansion in sensitive or bad-news indicators.*

2.4 | From Quantity to Composition: A Legitimacy-Theory Measurement Agenda

Taken together, prior research provides three pillars for a composition-focused analysis. First, legitimacy theory predicts that firms under pressure will disclose more but may do so strategically, privileging favorable indicators to maintain legitimacy (Deegan 2019; Ashforth and Gibbs 1990; Cho et al. 2010). Second, crisis settings amplify both the demand for information and managerial incentives to curate it, producing ambiguous predictions for substance (O'Donovan 2002; Maroun 2018). Third, the diffusion of corporate SR and the apparent plateau in incidence (KPMG 2024) underscore the need to move beyond volume to evaluate whether reporting is maturing in quality.

A composition approach operationalises these ideas by (i) coding indicators according to news direction (good/bad/neutral) and sensitivity (stakeholder-salient, reputationally risky), (ii) tracking changes in the disclosure mix across environmental

(EN), social (SO) and economic (EC) domains and (iii) comparing firms with differing exposure to environmental scrutiny. If firms are genuinely shifting toward substantive accountability, we expect growth not only in disclosed indicators but also in sensitive and bad-news coverage, especially in high-impact sectors subject to stronger scrutiny (Rodrigues and Mendes 2018). Conversely, an expanding emphasis on good/neutral indicators with little movement in sensitive areas would support a legitimacy-maintenance explanation (Melloni et al. 2017; Zharfpeykan 2021).

The literature collectively suggests that rising corporate SR incidence does not guarantee rising substance. Under legitimacy pressures, firms can increase disclosure while avoiding content that challenges narratives of responsibility, and crisis conditions may intensify both the demand for information and incentives for selective transparency. Consequently, assessing indicator composition—with particular attention to sensitive and bad-news categories—provides a more valid test of reporting maturity than volume metrics.

Building on legitimacy theory, two contrasting trajectories are possible. If SR is shifting toward substantive accountability, expansion should be accompanied by parallel growth in sensitive and adverse indicators. If, however, reporting primarily serves legitimacy maintenance, growth will be concentrated in good and non-sensitive categories, with limited movement in bad-news and sensitive areas (Ashforth and Gibbs 1990; Cho et al. 2010; Melloni et al. 2017). In this study, a shift toward substantive reporting is reflected in parallel growth in sensitive and bad-news indicators, whereas symbolic expansion is reflected in disproportionate growth in good and non-sensitive items. Accordingly, this study hypothesises that:

H3. *Growth in sustainability reporting is disproportionately concentrated in good and non-sensitive indicators rather than in bad-news and sensitive indicators.*

This hypothesis allows us to directly test whether recent reporting expansion reflects symbolic alignment or a shift toward substantive accountability.

This study advances this agenda through a longitudinal, indicator-level analysis of reporting strategies across crisis phases and environmental exposure. Together, these hypotheses frame the empirical analysis and allow us to assess whether recent expansions in SR reflect symbolic legitimacy management or substantive change.

3 | Method

3.1 | Sample Selection

To explore trends in reporting strategies, this study hand-collected and analysed sustainability report content¹ for 576 U.S.-listed firms from the Forbes Global 2000 over 2018–2023. We excluded 212 firms that did not use the Global Reporting Initiative (GRI) standards across the test period, yielding a final panel of 364 firms and 2184 firm-year reports: 158 high-impact firms (948 observations) and 206 low-impact firms (1236 observations),

following the environmental impact classification in Andersen and Bams (2022). The initial population consists exclusively of U.S.-listed Forbes Global 2000 firms, that is, very large and highly visible corporations. Restricting the sample to firms that consistently applied the GRI framework, therefore, differentiates among large corporations with broadly comparable resource bases and public exposure. Firms with inconsistent GRI use were excluded because the analysis relies on indicator-level composition scores derived from the GRI standards; partial or intermittent adoption would introduce structural incomparability and undermine cross-firm and longitudinal consistency.

The resulting dataset constitutes an unbalanced panel: firms are included if they produced at least one sustainability report during the study period, and firm-year observations are included only for years in which a report was published. Missing firm-year observations are not imputed. Accordingly, the reported sample size ($N=2184$) reflects the total number of observed firm-year reports included in the analysis.

To trace trends—and to assess whether the COVID-19 shock affected reporting strategies—we analysed two pre-pandemic years (2018–2019), two acute pandemic years (2020–2021) and the post-acute phase (2022–2023).²

GRI is among the most widely adopted frameworks and uniquely centres accountability for a company's economic, environmental and societal impacts (Adams and Abhayawansa 2022). Consistent with KPMG (2024), global corporate SR incidence is high ($\approx 79\%$ of N100; 96% of G250), with North America leading ($\approx 97\%$ of N100), and GRI remains the dominant standard. Focusing on large US firms (Forbes 2000) enhances comparability and data availability as larger firms possess the resources to produce more comprehensive reports, while smaller firms were more likely to prioritise economic survival during the acute phase of the pandemic (Klymenko and Lillebrygfjeld Halse 2022).

This study aims to identify firm changes in reporting strategies among the largest U.S. companies in response to evolving regulation and stakeholder demands, and to test whether the pandemic altered the substance and mix of GRI-aligned disclosures over time. Although firms in the sample differ in size, industry and business models, comparability is ensured in three ways. First, the sample is restricted to very large US-listed firms, limiting extreme variation in reporting capacity. Second, all firms are evaluated against the same standardised GRI indicator set. Third, the empirical analysis relies on firm fixed-effects models, so inferences are based on within-firm changes in reporting strategies over time rather than cross-sectional differences between firms.

3.2 | Content Analysis

Following Van der Laan (2009), we adopt a theory–measure fit by aligning our operationalisation with the nature of sustainability disclosure. To move beyond volume, we analyse both the breadth and the composition of reporting, that is, the news direction of indicators (positive/good, negative/bad, neutral) and their sensitivity (sensitive vs non-sensitive), to distinguish

greenwashing-leaning profiles from more representative strategies (Zharfpeykan 2021). Building on Zharfpeykan's (2021) legitimacy-theory classification, each GRI indicator is coded by: (i) news direction, where increases in performance reflect good (e.g., higher recycling rates), deterioration reflects bad (e.g., increased water withdrawal) and items without inherent valence (e.g., political contributions) are neutral; and (ii) sensitivity, which flags topics likely to attract public scrutiny, for example, legal breaches, harm or reputational risk—as sensitive, with other items classified as non-sensitive.

We apply this scheme to the GRI general indicators ($N=90$) across economic (EC), environmental (EN) and social (SO) categories, which are comparable across industries and enable meaningful cross-sectional analyses. A representative strategy—regardless of actual performance—would entail coverage not only of good, neutral and less sensitive areas but also of harmful and sensitive aspects central to stakeholder assessment (Zharfpeykan 2021). For each firm-year report, we conduct content analysis to determine whether each indicator is disclosed, recording binary presence/absence (1/0) at the indicator level. This indicator-level panel permits tests of whether firms' disclosure mix (not just counts) shifts over time in ways consistent with substantive accountability versus symbolic legitimacy management.

3.3 | Indicator Coding and Reliability

Building on the conceptual framework outlined above, this study operationalises and updates Zharfpeykan's (2021) legitimacy-theory classification by applying it to the current GRI general standards and implementing it at a comprehensive indicator level. This extension enables systematic cross-sectional and longitudinal analysis of reporting strategies using a standardised disclosure framework.

Coding was undertaken at the indicator level, prior to analysing any firm reports, to ensure that subsequent analyses reflect firms' disclosure choices. Each GRI indicator was classified along the two dimensions described in Section 3.2: news direction (good, bad, neutral) and sensitivity (sensitive vs non-sensitive).

Importantly, this study does not operationalise 'substantive' reporting using a fixed quantitative threshold (e.g., a minimum number or proportion of bad or sensitive indicators). Defining such a threshold would be inherently arbitrary and inconsistent with the study's legitimacy-theory framing. Instead, substantiveness is assessed relationally and contextually by examining patterns of indicator disclosure over time within the same firm and relative to peer firms operating in comparable industry and regulatory environments. Substantive reporting is therefore inferred from sustained broadening of disclosure into stakeholder-relevant, sensitive and adverse indicator categories, rather than from marginal or isolated increases in individual indicators. This approach treats reporting strategies as existing along a continuum from predominantly symbolic to more substantive, rather than as a binary classification.

The coding was conducted independently by two authors, both of whom have prior research experience in SR and content

analysis. Coders were trained using a detailed coding manual derived from legitimacy and greenwashing research. Coding was conducted blind to firm identity, industry classification and year. Inter-rater reliability was assessed using Krippendorff's alpha, which exceeded 0.80 across all categories, indicating strong agreement (Krippendorff 2009). Disagreements were resolved through discussion and consensus, and the coding framework was refined accordingly.

The final classification covers the full set of general GRI Economic, Environmental and Social indicators applicable across industries ($N=90$) under the latest version of the standards. Sector-specific indicators were excluded because they are not uniformly applicable and would confound cross-industry comparisons. Indicators were first classified by news direction: 'Good' where improvements clearly signal better performance (e.g., reductions in emissions, GRI 305-5), 'Bad' where higher levels reflect adverse impacts (e.g., water withdrawal, GRI 303-1) and 'Neutral' where more or less disclosure does not inherently imply better or worse performance (e.g., political contributions, GRI 415-1). Indicators were then classified as Sensitive when they relate to violations or harms likely to attract regulatory or stakeholder scrutiny, and Non-sensitive when they reflect routine activities unlikely to invite adverse attention (e.g., material use, GRI 301-1). For example, fines for non-compliance with environmental laws (GRI 307-1) were coded as Bad/Sensitive, emissions reductions (GRI 305-5) as Good/Sensitive and public policy engagement (GRI 415-1) as Neutral/Sensitive.

4 | Results

4.1 | Trends in Disclosure Volume and Industry Differences

Table 1 reports descriptive statistics for the average number of GRI indicators reported across the test period.

The initial analysis of the sample firms indicates that the average disclosure rose over the period, both in the total number of GRI indicators and within each category (EC, EN, SO). Yet, despite this upward trend, firms still reported fewer than half of the indicators specified by GRI—except for Social in 2023, which crossed that threshold. This section examines overall disclosure trends and whether they differ systematically by environmental exposure and across the COVID-19 period, directly testing the volume-based components of H1 and H2.

To assess whether restricting the sample to firms that consistently applied GRI materially altered its composition, we

compared the industry distribution of the full Forbes Global 2000 US subsample ($N=576$) with that of the final GRI-reporting panel ($N=364$). The distributions remain broadly similar across major sectors, with financially, environmentally and socially salient industries—including financial services, manufacturing, energy and construction, information technology and utilities—well represented in both samples. Importantly, no environmentally high-impact industry group drops out of the final panel. This suggests that the exclusion of non-GRI reporters did not materially reshape the sectoral profile of the sample, and that the trends documented below are unlikely to be driven by mechanical changes in industry composition rather than shifts in reporting strategies among large US firms.

To test whether these changes were statistically significant—and whether patterns differed by environmental exposure—we estimated firm fixed-effects panel models with standard errors clustered by firm. Following Andersen and Bams (2022), firms were classified as environmentally high-impact or low-impact to enable like-for-like comparisons of reporting strategies. This classification allows us to test H1, which predicts stronger growth in environmentally sensitive and adverse disclosure among high-impact firms, particularly within environmental categories.

The models track within-firm change over time in total GRI counts and by category (EC, EN, SO), control for time-invariant firm characteristics, and include year $\times$ impact interactions to capture differential trends between high- and low-impact industries. Table 2 reports the fixed-effects results.

The results indicate that the overall disclosure rose consistently and significantly across the period, as did category-level reporting (economic, environmental, social). The trend was strongest among high-impact firms, for whom all coefficients were positive and significant. Both industry groups recorded the largest gains in Social indicators, averaging $\beta=3.51$ (high-impact) and $\beta=3.15$ (low-impact) per year, that is, roughly three additional social indicators annually for each group. Comparing groups, high-impact industries showed larger increases across all three categories; however, the difference in trends between high- and low-impact firms was statistically significant only for Environmental indicators. In sum, GRI-aligned reporting improved broadly, with high-impact firms leading on environmental disclosure. Yet despite this upward trajectory, reported indicators in each category remain well below the full GRI-set. Consistent with H1, high-impact firms exhibit significantly stronger growth in environmental disclosure, indicating greater responsiveness to legitimacy pressures tied to environmental exposure.

TABLE 1 | Descriptive statistics.

	2018	2019	2020	2021	2022	2023
Average number of GRI indicators reported (out of 90)	14.01	19.42	25	31.34	35.51	36.71
Average number of EC indicators reported (out of 18)	2.25	2.98	3.8	4.94	5.68	6.3
Average number of EN indicators reported (out of 32)	5.8	7.9	9.97	12.47	14.12	12.8
Average number of SO indicators reported (out of 40)	5.96	8.5	11.24	13.93	15.73	24.26

TABLE 2 | Fixed effects regression results for the number of GRI-indicators reported across high- and low-impact industries.

GRI category	Environmentally high-impact industries		Environmentally low-impact industries		Trend × environmental impact ^a	
	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)
Total reporting (out of 90)	5.33*** (0.46)	(4.43, 6.24)	4.40*** (0.35)	(3.70, 5.10)	−0.93 (0.58)	(−2.07, 0.20)
<i>R</i> ² (within)	0.24		0.25		0.24	
Economic (out of 18)	0.89*** (0.08)	(0.73, 1.06)	0.80*** (0.07)	(0.67, 0.94)	−0.09 (0.11)	(−0.30, 0.12)
<i>R</i> ² (within)	0.20		0.21		0.21	
Environment (out of 32)	1.92*** (0.18)	(1.56, 2.28)	1.36*** (0.13)	(1.11, 1.61)	−0.56** (0.22)	(−1.00, −0.12)
<i>R</i> ² (within)	0.20		0.18		0.19	
Social (out of 40)	3.51*** (0.25)	(3.01, 4.01)	3.15*** (0.21)	(2.75, 3.56)	−0.36 (0.33)	(−1.00, 0.28)
<i>R</i> ² (within)	0.33		0.35		0.34	
<i>N</i> (obs)	158 (948)		206 (1236)		364 (2184)	

Note: The year 2018 is considered the base for the trend comparison. Standard errors are in parentheses.

^aThe results compare environmentally low-impact industries with high-impact industries. The sign of the coefficients indicates whether the disclosure rate was higher or lower in low-impact industries relative to high-impact industries.

***p* < 0.01.

****p* < 0.001.

To test whether the pandemic affected disclosure trends, we partitioned the sample into pre-COVID (2018–2019), acute COVID (2020–2021) and post-acute (2022–2023). We next examined whether the COVID-19 period was associated with shifts in disclosure patterns, as predicted in H2. Table 3 reports firm fixed-effects regressions comparing changes in the number of reported indicators for environmentally high-impact and low-impact industries from pre-COVID to acute COVID and acute COVID to post-acute.

From pre-COVID to acute COVID, total disclosure rose sharply in both groups: high-impact +13.46 indicators (SE = 1.44, *p* < 0.001) and low-impact +9.94 (SE = 1.01, *p* < 0.001). From acute COVID to post-acute, growth slowed: high-impact +8.20, low-impact +7.74; the between-group difference (−1.99, SE = 1.18) was not statistically significant. By category, Economic gains were modest and similar across groups. In Environmental, high-impact firms outpaced low-impact (pre to acute COVID: +5.54 vs. +3.46; acute COVID to post-acute: +2.42 vs. +2.10) and the change in trends (−1.20, SE = 0.46) showed a significantly smaller increase for low-impact industries (*p* < 0.05). Social was the only category with accelerating growth in the post-acute phase (high-impact +6.09 then +7.50; low-impact +4.79 then +7.34); the group difference (−0.73, SE = 0.65) was not statistically significant.

Overall, indicator counts improved for both groups, but economic and environmental momentum decelerated, while social momentum continued to climb, suggesting a broadly consistent shift in reporting priorities. The post-acute slowdown may signal either a business-as-usual plateau or incipient stagnation. This pattern is consistent with H2: the acute COVID period

coincides with a relative reweighting toward social disclosure, without a proportional expansion in adverse or sensitive categories. Because counts alone cannot reveal strategy, we next coded reports by GRI coverage to examine composition, that is, which areas firms chose to disclose—tracking within-period shifts and any discrete changes across the three phases.

4.2 | Corporate Sustainability Reporting Strategies

A representative strategy is characterised by broader coverage across news-direction and sensitivity categories—especially a willingness to disclose bad and/or sensitive indicators in domains of high relevance (e.g., environmental impact for high-impact industries). By contrast, a greenwashing strategy typically limits negative/sensitive content and prioritises positive, neutral or non-sensitive topics—regardless of the total number of indicators reported. Classic greenwashing thus presents high-level positives and non-sensitive issues while giving minimal attention to adverse or sensitive disclosures. These tendencies are relative rather than absolute. To test H3, which concerns whether disclosure growth is concentrated in symbolic versus substantive categories, we analyse the composition of reporting across news direction and sensitivity.

Using a legitimacy-theory lens, we examine the composition of disclosure by analysing the average number of GRI indicators reported in each category across the test period. Table 4 reports, for Economic, Environmental and Social domains, both (i) total indicators and (ii) the proportion of each news × sensitivity sub-category: (1) good/sensitive, (2) good/not-sensitive, (3) bad/sensitive, (4) neutral/sensitive, (5) neutral/not-sensitive.

TABLE 3 | Fixed-effects regression results for the number of GRI indicators reported by high- and low-impact industries across pre-COVID, acute COVID and post-acute phases.

GRI indicators reported	Environmentally high-impact industries				Environmentally low-impact industries				Trend × environmental impact ^a (low-high)	
	Pre-COVID/acute COVID		Acute COVID/post-acute		Pre-COVID/Acute COVID		Acute COVID/post-acute		Coefficient (SE)	(95% conf. interval)
	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)		
Total reporting	13.46*** (1.44)	(10.62, 16.30)	8.20*** (1.55)	(5.13, 11.27)	9.94*** (1.01)	(7.95, 11.93)	7.74*** (1.15)	(5.47, 10.01)	−1.99* (1.18)	(−4.32, 0.34)
Economic	1.83*** (0.27)	(1.30, 2.36)	1.75*** (0.31)	(1.15, 2.36)	1.70*** (0.20)	(1.30, 2.09)	1.5*** (0.22)	(1.07, 1.95)	−0.19 (0.22)	(−0.62, 0.25)
Environment	5.54*** (0.061)	(4.34, 6.74)	2.42*** (0.59)	(1.26, 3.59)	3.46*** (0.37)	(2.72, 4.19)	2.10*** (0.41)	(1.29, 2.91)	−1.20*** (0.46)	(−2.12, −0.29)
Social	6.09*** (0.65)	(4.81, 7.37)	7.5*** (0.86)	(5.80, 9.20)	4.79*** (0.49)	(3.82, 5.76)	7.34*** (0.68)	(5.99, 8.69)	−0.73 (0.65)	(−2.00, 0.54)
N (obs)	632 (158)				206 (824)				364 (2184)	

Note: The year 2018 is considered the base for the trend comparison. Standard errors are in parentheses.

^aThe results compare environmentally low-impact industries with high-impact industries. The sign of the coefficients indicates whether the disclosure rate was higher or lower in lower-impact industries relative to high-impact industries.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

As shown in Table 4, the average number of indicators reported rose across the period for all categories, except Social: Neutral/Not-Sensitive. Even so, absolute levels remain well below the full GRI set. The Economic share of total disclosure stayed consistently under the GRI benchmark (20%), with reporting concentrated in Good/Sensitive items. By contrast, the Environmental share was generally above the 36% GRI benchmark for most years; however, firms placed their greatest emphasis on Bad/Not-Sensitive indicators, while Bad/Sensitive coverage remained below half of the indicators available in GRI. Social indicators were relatively stable in proportion, except in 2023, when a sharp increase in social disclosure coincided with a decline in environmental coverage. Within Social, firms prioritised Good/Sensitive items, while Bad/Sensitive coverage remained broadly flat. This 2023 shift suggests a reallocation of reporting attention rather than an overall contraction, as total disclosure continued to rise. One plausible explanation is that firms increasingly prioritised social topics—such as employee welfare, health and safety, diversity and workforce practices—in the post-acute phase, leading to rapid expansion in Social indicators. At the same time, Environmental reporting, which had risen steadily through 2022 and was already the most developed category, appears to have stabilised, with fewer new indicators added. Taken together, the 2023 pattern points to a shift in perceived legitimacy priorities rather than a retreat from environmental reporting.

These results highlight why composition matters; even as total Economic, Environmental and Social counts trend upward, the distribution across news-direction × sensitivity reveals shifting reporting priorities. This emphasis on composition echoes Clarkson et al. (2008), who show that socio-political theories are more powerful in explaining the nature and tone of discretionary disclosures than their sheer volume. Consistent with their evidence, our results indicate that firms expand reporting while continuing to privilege less sensitive and more favourable indicators, reinforcing the importance of examining what is disclosed rather than how much. Consistent with H3, growth is disproportionately concentrated in good and non-sensitive indicators, while expansion in bad-news and sensitive categories remains limited.

To probe strategy more directly, we partition firms into high- and low-environmental-impact groups. Figure 1 traces how each group's reporting strategy evolved relative to GRI benchmarks, offering a clearer view of these strategies and trends compared to the GRI standards and whether changes reflect representative coverage or selective (greenwashing-leaning) emphasis.

Figure 1 shows that, in 2018, both industry groups displayed similarly low overall disclosure and a greenwashing-leaning profile: limited attention to economic/environmental/social (E/EN/SO) Bad-Sensitive items, emphasising good and neutral categories. For low-impact industries, this pattern persisted, and while total disclosure rose, the ratio devoted to Sensitive and Bad categories did not improve. By contrast, high-impact industries showed a modestly different trajectory. While continuing to disclose economic and environmental good-sensitive indicators, they increased coverage in environmental bad-not-sensitive and social good-sensitive and bad-sensitive categories as overall reporting grew. By 2023 (post-acute phase), high-impact firms

TABLE 4 | Average level of disclosure under different news direction and sensitivity categories.

News direction and sensitivity	Number of indicators in the GRI standards		2018		2019		2020		2021		2022		2023	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Total economic	18	20	2.25	16	2.98	15	3.8	15	4.94	16	5.68	16	6.3	17
Good/Sensitive	2	2	1.08	8	1.35	7	1.71	7	2.06	7	2.29	6	2.36	6
Good/Not sensitive	6	7	0.88	6	1.17	6	1.41	6	1.73	6	1.93	5	2.08	6
Bad/Sensitive	2	2	0	0	0.04	0	0.11	0	0.27	1	0.36	1	0.43	1
Neutral/Sensitive	3	3	0.28	2	0.38	2	0.48	2	0.61	2	0.72	2	0.81	2
Neutral/Not sensitive	5	6	0	0	0.04	0	0.09	0	0.27	1	0.37	1	0.43	1
Total environment ^a	32	36	5.8	41	7.9	41	9.97	40	12.47	40	14.12	40	12.8	35
Good/Not sensitive	8	9	1.13	8	1.49	8	1.82	7	2.15	7	2.46	7	2.55	7
Bad/Sensitive	19	21	1.45	10	1.93	10	2.39	10	2.91	9	3.28	9	3.37	9
Bad/Not sensitive	5	6	3.21	23	4.5	23	5.75	23	7.4	24	8.38	24	8.8	24
Total social	40	44	5.96	43	8.5	44	11.24	45	13.93	44	15.73	44	24.26	66
Good/Sensitive	1	1	3.33	24	4.88	25	6.41	26	8	26	8.88	25	9.01	25
Good/Not sensitive	20	22	0.04	0	0.12	1	0.19	1	0.31	1	0.41	1	0.43	1
Bad/Sensitive	16	18	1.98	14	2.67	14	3.54	14	4.34	14	5.18	15	5.21	14
Neutral/Sensitive	1	1	0.47	3	0.65	3	0.86	3	1.05	3	1.13	3	1.16	3
Neutral/Not sensitive	2	2	0.14	1	0.18	1	0.22	1	0.23	1	0.12	0	0.06	0
Total reporting	90	100	14.01	100	19.42	100	25	100	31.34	100	35.51	100	36.71	100

^aNo indicators were coded as Environmental Good/Sensitive or Neutral disclosures.

disclosed, on average, about 40% of social bad-sensitive and 55% of environmental bad-sensitive indicators relative to the GRI set—evidence of movement toward a more representative strategy. Nonetheless, even these firms fell short of comprehensive coverage in sensitive areas, underscoring that the shift toward substantive transparency remains partial and more pronounced in high- than low-impact sectors. This provides partial support for H1: environmentally high-impact firms move further toward sensitive environmental disclosure but still fall short of comprehensive coverage.

This pattern closely reflects the symbolic-substantive distinction articulated by Ashforth and Gibbs (1990) and applied to corporate SR by Cho et al. (2010) and Melloni et al. (2017). While high-impact firms show some movement toward more

representative coverage—particularly in environmentally sensitive disclosures—the continued under-representation of bad and sensitive indicators suggests that legitimacy maintenance remains a dominant reporting logic. The persistence of selective emphasis mirrors earlier findings that firms expand disclosure while avoiding the most reputationally risky content (Hahn and Lülfs 2014; Zharfpeykan 2021).

To test whether changes in the number of reported indicators within each category were statistically significant over time, we estimated firm fixed-effects panel models that control for time-invariant firm characteristics and isolate within-firm variation. Standard errors were clustered at the firm level to address serial correlation. Because this analysis examines multiple disclosure categories simultaneously, we control for multiple testing using

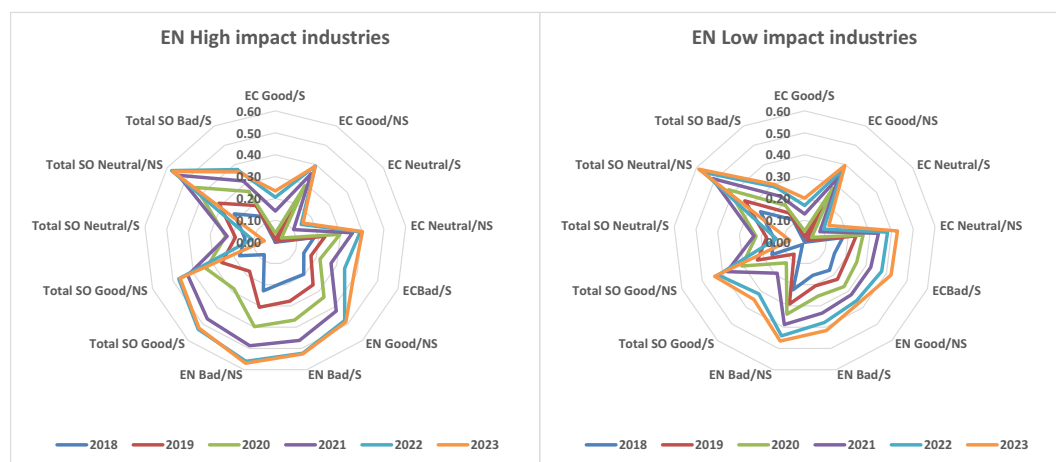


FIGURE 1 | Reporting strategies across high-impact and low-impact environmental industries.

the Benjamini–Hochberg false discovery rate (BH-FDR) procedure for the Trend $\times$ Environmental-impact interaction effects. Results are presented in Table 5.

Table 5 shows that the number of reported indicators increased significantly over time across almost all categories for both high- and low-impact industries. The only exception was social: neutral/sensitive, which declined for both groups. One plausible interpretation is that firms shifted away from neutral discussion of socially sensitive issues toward either more positively framed disclosures or more explicitly evaluative reporting (bad/sensitive), as topics such as employee welfare, safety and diversity became more salient. In this sense, the decline in neutral/sensitive indicators appears to reflect changing disclosure framing rather than reduced attention to social issues. Within the economic category, the largest gains occurred in good/not-sensitive and neutral/not-sensitive indicators. In the environmental category, the most pronounced growth was in bad/sensitive indicators. For the social category, the strongest increase was in good/not-sensitive indicators.

Although overall patterns were broadly similar across groups, evidence of differential growth rates should be interpreted cautiously in light of multiple testing. After applying BH-FDR correction, statistically reliable between-group differences are concentrated in a limited subset of outcomes, primarily within Environmental disclosure categories. In particular, high-impact industries exhibit significantly steeper growth in environmental bad/not-sensitive, environmental bad/sensitive and environmental good/not-sensitive indicators. A statistically robust between-group difference is also observed for total social good/sensitive indicators. For the remaining categories, interaction effects do not remain statistically reliable once controlling for false discoveries.

The magnitudes of these effects are substantively meaningful. Translating the estimated trends into cumulative changes over the sample period indicates that environmental bad/sensitive disclosures increased by approximately 6.9 indicators among high-impact firms, compared to about 5.1 among low-impact firms, representing roughly 36% and 27% of the 19-indicator

benchmark, respectively. These findings indicate that observed differences reflect economically meaningful shifts in disclosure composition rather than marginal reporting changes.

For categories where between-group differences are not statistically robust, the similarity in trends suggests that environmentally high- and low-impact firms followed broadly comparable disclosure trajectories. One plausible interpretation is that SR has become increasingly institutionalised, with firms converging on comparable GRI-aligned practices regardless of environmental exposure. This convergence is consistent with legitimacy theory: once reporting becomes an expected organisational response, firms across industries expand disclosure in parallel, even if their underlying risk profiles differ. Prior research similarly finds that SR diffuses in a standardised manner across sectors, producing limited cross-industry variation outside environmentally salient domains (Cho et al. 2010; O'Donovan 2002; Zharfpeykan and Akroyd 2023). In our data, meaningful between-group differences emerge selectively and are concentrated primarily in environmental sub-categories, where legitimacy pressures are most directly tied to industry impact. Overall, these trends indicate widening disclosure coverage, but with growth concentrated in specific news-direction $\times$ sensitivity cells rather than uniformly across the full GRI set.

To assess whether the exogenous COVID-19 shock altered corporate SR strategies, we partition the sample into pre-COVID (2018–2019), acute COVID (2020–2021) and post-acute (2022–2023) periods and estimate fixed-effects models with phase-specific time trends. Table 6 reports the results and compares disclosure growth between environmentally high-impact and low-impact industries. Statistical inference for between-group differences is again based on BH-FDR adjusted q -values for the phase-specific trend $\times$ environmental-impact terms.

Across both industry groups, reporting activity increased significantly in almost all disclosure categories over the full sample period, indicating a broad expansion in SR. However, growth patterns differ across phases. For both high- and low-impact firms, disclosure growth generally decelerates in the acute COVID to post-acute period, particularly within Environmental

TABLE 5 | Fixed-effects regression results: Analysing changes in reporting strategies by number of reported indicators.

GRI category	Environmentally high-impact industries		Environmentally low-impact industries		Trend × environmental impact		
	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	q (BH-FDR)
Economic							
Good/Sensitive	0.11*** (0.01)	(0.07, 0.13)	0.09*** (0.01)	(0.06, 0.11)	−0.02 (0.02)	(−0.05, 0.02)	0.53
Good/Not sensitive	0.27*** (0.04)	(0.02, 0.05)	0.27*** (0.03)	(0.22, 0.33)	0.003 (0.05)	(−0.09, 0.09)	0.94
Bad/Sensitive	0.10*** (0.01)	(0.07, 0.13)	0.11*** (0.01)	(0.09, 0.14)	0.01 (0.02)	(−0.03, 0.05)	0.81
Neutral/Sensitive	0.11*** (0.01)	(0.08, 0.13)	0.08*** (0.01)	(0.06, 0.1)	−0.02 (0.02)	(−0.05, 0.01)	0.41
Neutral/Not sensitive	0.24*** (0.03)	(0.17, 0.30)	0.25*** (0.03)	(0.20, 0.31)	0.02 (0.04)	(−0.07, 0.10)	0.82
Environment							
Good/Not sensitive	0.50*** (0.05)	(0.40, 0.60)	0.33*** (0.03)	(0.26, 0.40)	−0.17 (0.06)	(−0.29, −0.05)	0.03**
Bad/Sensitive	1.37*** (0.11)	(1.14, 1.60)	1.03*** (0.08)	(0.87, 1.19)	−0.34 (0.14)	(−0.62, −0.06)	0.05**
Bad/Not sensitive	0.36*** (0.03)	(0.30, 0.43)	0.24*** (0.03)	(0.19, 0.30)	−0.12 (0.04)	(−0.21, −0.04)	0.03**
Social							
Good/Sensitive	0.10*** (0.008)	(0.08, 0.12)	0.07*** (0.007)	(0.06, 0.08)	−0.03 (0.01)	(−0.05, −0.008)	0.03**
Good/Not sensitive	1.24*** (0.11)	(1.02, 1.46)	1.17*** (0.09)	(0.99, 1.35)	−0.07 (0.14)	(−0.36, 0.21)	0.81
Bad/Sensitive	0.80*** (0.08)	(0.64, 0.96)	0.62*** (0.06)	(0.49, 0.75)	−0.18 (0.10)	(−0.39, 0.02)	0.20
Neutral/Sensitive	−0.02*** (0.007)	(−0.03, −0.006)	−0.01*** (0.005)	(−0.02, −0.003)	0.0005 (0.009)	(−0.01, 0.02)	0.81
Neutral/Not sensitive	0.15*** (0.02)	(0.12, 0.18)	0.14*** (0.01)	(0.11, 0.17)	−0.006 (0.02)	(−0.05, 0.03)	0.82
N (obs)	158 (948)		206 (1236)		364 (2184)		

Note: The dependent variables are the number of GRI indicators reported within each news-direction × sensitivity category. All models include firm fixed effects and a linear time trend (2018 = 0). 'Environmentally high-impact' and 'Environmentally low-impact' coefficients report within-group time trends. Standard errors clustered at the firm level are in parentheses. The interaction term (Trend × environmental impact) captures differences in growth rates between environmentally low-impact and high-impact industries. Negative coefficients indicate steeper growth among high-impact firms. Because multiple interaction effects are tested simultaneously, statistical inference for between-group differences is based on Benjamini–Hochberg false discovery rate-adjusted q -values reported in the final column. Following the literature, results with $q \leq 0.10$ are interpreted as statistically robust.

** $p < 0.01$.

*** $p < 0.001$ denote conventional significance levels for within-group trends.

and Social categories and for economically non-sensitive indicators. This pattern suggests that although the pandemic coincided with a surge in reporting, post-acute momentum weakened, especially in areas most closely associated with sensitive or adverse information.

Consistent with H1, statistically robust differences between high- and low-impact industries are concentrated in environmental disclosure categories during the pre-COVID to acute COVID period. After controlling for multiple testing using the Benjamini–Hochberg false discovery rate procedure, environmentally high-impact industries exhibit significantly stronger growth than low-impact industries in environmental bad-news and sensitive indicators ($q \leq 0.10$), as well as in closely

related environmental sub-categories (all $q \leq 0.10$). These results indicate that firms operating in environmentally exposed sectors intensified disclosure, particularly, in those environmental areas most salient to legitimacy concerns at the onset of the crisis.

In contrast, no between-group differences in disclosure trends remain statistically robust in the acute COVID to post-acute period after FDR correction (all $q > 0.10$). This finding indicates that the initial divergence in environmental disclosure behaviour between high- and low-impact industries does not persist into the post-acute phase once acute crisis pressures subside, suggesting a partial convergence in reporting trajectories once acute crisis pressures subside.

TABLE 6 | Fixed-effects regression results: Analysing changes in reporting strategies by number of reported indicators across pre-COVID, acute COVID and post-acute phases.

GRI category	Environmentally high-impact industries				Environmentally low-impact industries				Trend × environmental impact ^a			
	Pre-COVID/Acute COVID		Acute COVID/post-acute		Pre-COVID/Acute COVID		Acute COVID/post-acute		Pre-COVID/Acute COVID		Acute COVID/post-acute	
	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)	Coefficient (SE)	(95% conf. interval)
Economic												
Good/Sensitive	0.17*** (0.04)	(0.10, 0.24)	0.26*** (0.05)	(0.16, 0.35)	0.15*** (0.03)	(0.10, 0.21)	0.19*** (0.04)	(0.11, 0.26)	0.009 (0.009)	(-0.04, 0.05)	0.82 (0.09)	(-0.03, 0.74)
Good/Not sensitive	0.68*** (0.12)	(0.45, 0.91)	0.42*** (0.12)	(0.18, 0.66)	0.66*** (0.09)	(0.49, 0.84)	0.45*** (0.09)	(0.28, 0.62)	0.02 (0.02)	(-0.11, 0.15)	0.81 (0.12)	(-0.17, 1)
Bad/Sensitive	0.18*** (0.04)	(0.09, 0.26)	0.23*** (0.05)	(0.13, 0.33)	0.23*** (0.04)	(0.15, 0.30)	0.22*** (0.04)	(0.14, 0.30)	-0.03 (0.03)	(-0.08, 0.02)	0.56 (0.062)	(-0.056, 1)
Neutral/Sensitive	0.19*** (0.04)	(0.12, 0.26)	0.24*** (0.05)	(0.15, 0.33)	0.16*** (0.03)	(0.11, 0.21)	0.17*** (0.03)	(0.11, 0.24)	0.02 (0.02)	(-0.03, 0.06)	0.68 (0.08)	(-0.02, 0.74)
Neutral/Not sensitive	0.61*** (0.10)	(0.41, 0.81)	0.35*** (0.11)	(0.12, 0.57)	0.49*** (0.08)	(0.34, 0.64)	0.5*** (0.09)	(0.32, 0.67)	0.04 (0.04)	(-0.08, 0.16)	0.70 (0.05)	(-0.22, 0.97)
Environment												
Good/Not sensitive	1.20*** (0.15)	(0.89, 1.50)	0.84*** (0.16)	(0.52, 1.15)	0.78*** (0.10)	(0.57, 0.98)	0.55*** (0.11)	(0.33, 0.76)	0.20** (0.03)	(0.03, 0.37)	0.05*** (0.03)	(-0.04, 1)
Bad/Sensitive	3.47*** (0.37)	(2.74, 4.20)	2.08*** (0.38)	(1.34, 2.83)	2.15*** (0.22)	(1.72, 2.58)	1.95*** (0.26)	(1.43, 2.47)	0.64*** (0.03)	(0.24, 1.03)	0.01*** (0.03)	(-0.42, 0.95)
Bad/Not sensitive	0.87*** (0.11)	(0.65, 1.10)	0.61*** (0.10)	(0.41, 0.82)	0.53*** (0.08)	(0.38, 0.68)	0.45*** (0.09)	(0.28, 0.62)	0.16*** (0.03)	(0.04, 0.29)	0.05*** (0.03)	(-0.05, 0.88)
Social												
Good/Sensitive	0.25*** (0.03)	(0.19, 0.31)	0.15*** (0.03)	(0.09, 0.21)	0.11*** (0.02)	(0.08, 0.15)	0.17*** (0.03)	(0.12, 0.23)	0.07*** (0.03)	(0.04, 0.10)	0.000*** (0.03)	(-0.04, 0.95)
Good/Not sensitive	3.29*** (0.35)	(2.60, 3.98)	1.71*** (0.38)	(0.96, 2.46)	2.96*** (0.29)	(2.38, 3.53)	1.75*** (0.31)	(1.14, 2.37)	0.20 (0.03)	(-0.22, 0.62)	0.57 (0.03)	(-0.49, 1)
Bad/Sensitive	2.04*** (0.25)	(1.54, 2.55)	1.27*** (0.28)	(0.71, 1.84)	1.29*** (0.18)	(0.84, 1.63)	1.24*** (0.22)	(0.80, 1.67)	0.35*** (0.03)	(0.07, 0.64)	0.05*** (0.03)	(-0.34, 1)
Neutral/Sensitive	0.06*** (0.02)	(0.02, 0.11)	-0.15*** (0.03)	(-0.20, -0.09)	0.07*** (0.02)	(0.03, 0.11)	-0.13*** (0.02)	(-0.17, -0.08)	-0.003 (0.03)	(-0.03, 0.02)	0.81 (0.02)	(-0.04, 1)
Neutral/Not sensitive	0.44*** (0.05)	(0.33, 0.54)	0.18*** (0.05)	(0.06, 0.27)	0.36*** (0.04)	(0.27, 0.44)	0.21*** (0.04)	(0.13, 0.30)	0.03 (0.03)	(-0.03, 0.09)	0.56 (0.04)	(-0.09, 1)
N(obs)		632 (158)				824 (206)				1456 (364)		1456 (364)

Note: Standard errors are in parentheses.

^aThe results compare environmentally low-impact industries with high-impact industries. The sign of the coefficients indicates whether the disclosure rate was higher or lower in low-impact industries relative to high-impact industries.

*** $p < 0.01$.

**** $p > 0.001$.

Taken together, these results provide qualified support for H1: environmentally high-impact firms exhibit stronger growth in environmentally sensitive and bad-news disclosures, but this pattern is concentrated in the pre-COVID/acute COVID phase. With respect to H2 and H3, although overall disclosure volumes continue to rise, post-acute growth is weaker and disproportionately concentrated in good and non-sensitive categories across both industry groups, while sensitive and bad-news indicators expand more slowly or stagnate. This compositional shift suggests that COVID-19 increased the scale of SR without producing a commensurate expansion in substantively demanding disclosures.

Interpreted through the crisis and legitimacy literature, this pattern is consistent with the view that exogenous shocks intensify incentives to communicate sustainability engagement but do not necessarily lead to deeper accountability (O'Donovan 2002; Maroun 2018). COVID-19 appears to have triggered an expansion in disclosure activity, particularly among environmentally exposed firms, yet the post-acute deceleration in sensitive and bad-news indicators indicates that firms primarily adjusted the volume and emphasis of reporting rather than its substantive balance. This distinction is especially relevant as firms transition toward more stringent regimes under the ISSB's IFRS Sustainability Disclosure Standards³ and the EU Corporate Sustainability Reporting Directive (CSRD),⁴ where regulatory and stakeholder expectations increasingly emphasise not only more disclosure but more representative and decision-useful coverage of material sustainability impacts.

4.3 | Internal and External Validation of the Composition Measure

These results provide systematic evidence supporting both the internal and external validity of the composition-based measure developed in this study.

Internal validity is supported by the coherence, stability and theoretical consistency of the observed patterns. Across the sample, firms did not expand disclosure uniformly but re-weighted their reporting mix in structured ways: good and non-sensitive indicators dominate, sensitive and bad-news indicators grow more slowly and environmentally sensitive categories expand more strongly in high-impact industries. These regularities are not mechanical consequences of the GRI framework; rather, they reflect selective and interpretable disclosure choices that align closely with legitimacy theory. The measure consistently differentiates disclosure profiles across domains, news direction, sensitivity, industry exposure and crisis phases, indicating that it captures meaningful variation in reporting strategies rather than noise. The concentration of between-group differences in environmental sub-categories, alongside convergence in economic and social areas, further supports the internal coherence of the framework and its ability to discriminate where legitimacy pressures are most salient.

External validity is supported in two ways. First, the measure behaves as predicted by prior literature: environmentally high-impact industries exhibit significantly greater coverage of

environmentally sensitive indicators, while firms overall privilege favourable and less sensitive disclosures—patterns documented in legitimacy and greenwashing research. Second, we construct environmental and social composition scores based on the proportion of GRI indicators disclosed within each news $\times$ sensitivity category, scaled by the total number of available GRI indicators. We then correlate these scores with the environmental and social pillar disclosure scores from Refinitiv Asset4, which provide independent assessments of firms' sustainability disclosure and performance. The correlations are positive and statistically significant (EN: $r=0.27$, $p<0.01$; SO: $r=0.30$, $p<0.01$). Given that Asset4 pillar scores incorporate a broad set of external evaluations of sustainability disclosure and performance, these associations indicate that firms exhibiting more extensive and substantive indicator coverage under our framework are also assessed more favourably by an established third-party provider.

Taken together, these results support the reliability and construct validity of the coding scheme and composition measure. They indicate that the framework not only produces stable and interpretable disclosure profiles but also aligns with both theoretical expectations and independent sustainability assessments, reinforcing its suitability for analysing corporate SR strategies.

5 | Discussions and Conclusions

This study set out to determine whether recent growth in corporate SR reflects substantive change or largely legitimacy maintenance, with a deliberate shift from counting disclosure volume to examining the composition of indicators disclosed. Using a hand-collected dataset, the study analysed sustainability reports from 576 US-listed firms selected from the Forbes 2000 list between 2018 and 2022. Building on the classification developed by Zharfpeykan (2021), which classifies corporate SR strategies through the lens of legitimacy theory, this study refines and updates the framework in line with the latest GRI standards. Framed by legitimacy theory, our analysis shows that while disclosures expanded across the period studied, firms—especially those in low-impact industries—continue to curate what they report, emphasising good and non-sensitive items and under-representing sensitive or 'bad-news' indicators. In short, expansion did not automatically translate into more complete engagement with material impacts. This result highlights that counting more indicators risks mistaking symbolic alignment for substantive accountability (Ashforth and Gibbs 1990; Deegan 2019; Cho et al. 2010; O'Donovan 2002). This evidence directly extends prior empirical work showing that disclosure growth does not necessarily translate into improved reporting quality (Hahn and Lölfs 2014; Zharfpeykan and Akroyd 2023). By applying a composition-based measure, our findings operationalise the legitimacy-theory concern raised by Deegan (2019) and Cho et al. (2010), that firms can respond to pressure by expanding disclosure while continuing to manage impressions through selective content.

Taken together, these findings provide support for the legitimacy-based hypotheses developed in Section 2. High-impact industries exhibit stronger growth in environmentally sensitive and

adverse indicators, particularly during the pre-COVID to acute COVID phase, consistent with heightened legitimacy pressures (qualified support for H1). The acute COVID period is associated with a pronounced shift toward social disclosure without a proportional expansion in sensitive or bad-news categories (supporting H2). More broadly, growth in SR remains disproportionately concentrated in good and non-sensitive indicators, with limited broadening into adverse and sensitive areas (supporting H3).

The results show that the largest gains occurred in Social indicators, followed by Environmental and Economic categories. High-impact industries disclosed more overall and held a statistically significant lead in environmental topics—consistent with stronger coercive and normative pressures where exposure and scrutiny are higher (Rodrigues and Mendes 2018; Deegan 2019). Yet even in these sectors, coverage remains well below GRI benchmarks, particularly for sensitive areas most salient to stakeholders. For instance, in 2023 (post-acute phase) firms covered, on average, only 53% of environmental bad or sensitive indicators. In low-impact industries, increases in total disclosure (from an average of 13 indicators in 2018 to 25 indicators in 2023) were not matched by a broadening into sensitive topics. For example, the reporting of sensitive social indicators increased only modestly, from an average of two indicators in 2018 to 3.7 indicators in 2023. This pattern indicates strategically selective reporting, consistent with prior research (Hahn and Lülfs, 2014; Melloni et al. 2017; Zharfpeykan 2021). These patterns mirror earlier evidence that firms can expand reporting while leaving material gaps intact (Zharfpeykan and Akroyd 2023; KPMG 2024).

A central implication of legitimacy theory is that organisations can satisfy rising expectations by curating the disclosure mix rather than transforming practices (Ashforth and Gibbs 1990; Deegan 2019). Our composition analysis shows a persistent emphasis on good/neutral indicators, with only modest growth in sensitive and negative items—precisely the disclosures that differentiate substance from symbolic communication (Cho et al. 2010; Maroun 2018). Where progress did occur, for example, small increases in environmental ‘bad/not-sensitive’ and social good/bad-sensitive items in high-impact industries—the gains remain incremental and below benchmark levels. In short, volume-based metrics can create an illusion of progress; without examining which indicators expand, it is difficult to judge whether reporting is genuinely becoming more decision-useful. These findings closely align with Clarkson et al. (2008), who show that while superior performers disclose more ‘hard’ information, socio-political pressures are particularly visible in the strategic use of softer, legitimising disclosures—supporting our interpretation that contemporary SR continues to blend accountability with impression management. They are also consistent with the greenwashing literature. Delmas and Burbano (2011) describe greenwashing as positive communication decoupled from underlying impacts, while Lyon and Montgomery (2015) emphasise selective disclosure as a central mechanism. The persistent emphasis on good and neutral indicators we document, alongside limited growth in sensitive and adverse disclosures, fits squarely with these accounts of strategic greenwashing.

Treating COVID-19 as an exogenous shock (Coombs 2007; Zharfpeykan and Ng 2021) allowed us to test whether reporting

strategies were sufficiently mature to broaden substantively under pressure. Disclosures continued to grow during and after the pandemic; however, the rate of improvement slowed post-acute phase in Economic and Environmental categories, while Social maintained an upward trajectory, and between-industry differences largely disappeared once acute crisis pressures subsided (He and Harris 2020; Parker and Narayanan 2023). The net effect was a bounded adjustment rather than a step-change toward fuller sensitive coverage, especially in low-impact industries. By contrast, longer-running climate pressures appear more enduring, with the largest increases in environmentally sensitive areas such as emissions (Markard and Rosenbloom 2020; Deutsche 2020; Tol 2018). Together, these results nuance the reported plateau in reporting incidence (KPMG 2024): reporting has not stalled, but quality-relevant improvements in composition have decelerated. In this sense, COVID-19 functioned less as a catalyst for substantive accountability and more as a stress test of existing reporting strategies. Consistent with Coombs’ (2007) victim-cluster framing, firms increased disclosure activity, particularly on social topics salient during the crisis, but did not systematically broaden sensitive environmental or economic coverage. This supports recent concerns that crisis-driven reporting responses may be adaptive yet bounded, reinforcing rather than transforming underlying legitimacy practices (Parker and Narayanan 2023; Zharfpeykan and Ng 2021).

This research makes several contributions to the literature on corporate SR strategies. First, it introduces a practical, indicator-level approach—coding news (good/bad/neutral) and sensitivity—that enables researchers and practitioners to distinguish symbolic from substantive reporting, moving the debate beyond disclosure volume (Cho et al. 2010; KPMG 2024; Zharfpeykan 2021). Methodologically, the study extends Zharfpeykan (2021) by translating a legitimacy-based disclosure framework into a replicable, indicator-level classification aligned with current GRI standards, enabling systematic analysis of disclosure composition across firms, industries and time. Second, by comparing high- and low-impact industries over time, the study shows how legitimacy pressures shape the mix of indicators: stronger scrutiny is associated with greater environmental coverage, yet sensitive topics remain under-reported—evidence of curated composition rather than uniform substantive change (Deegan 2019; Rodrigues and Mendes 2018; Zharfpeykan and Akroyd 2023). Third, using COVID-19 as a ‘victim-cluster’ crisis (Coombs 2007; Zharfpeykan and Ng 2021), the analysis distinguishes temporary, bounded shifts (e.g., social emphasis during the crisis) from durable changes in indicator composition. The limited broadening into sensitive categories suggests that crisis-induced pressures alone did not drive a structural move toward more substantive transparency (Parker and Narayanan 2023; Phang et al. 2023). By centring on indicator composition under legitimacy pressures, this study also offers a diagnostic lens that complements evolving reporting regimes such as the ISSB standards and the CSRD, which require transparent explanation of material topics and impacts. In practice, the framework helps firms and users assess whether expanded reporting reflects substantive accountability or selective transparency. More broadly, the study responds to calls to move SR research beyond whether firms disclose toward how and what they

disclose, thereby providing an empirically grounded way to test legitimacy theory in contemporary reporting contexts.

This study offers several practical contributions for both managers and stakeholders. For regulators and standard-setters, incidence and page counts are inadequate proxies for reporting quality. Our findings gain added relevance in light of recent regulatory developments, including the ISSB's standards and the CSRD, both of which signal a shift toward more rigorous, decision-useful reporting requirements. These frameworks emphasise materiality, connectivity with financial information and comprehensive coverage of sustainability matters—criteria that simple volume metrics cannot capture. Accordingly, guidance and assurance that explicitly target coverage of sensitive indicators and the balance of good/neutral versus negative items would better align disclosure with stakeholder needs (KPMG 2024; Bellucci et al. 2021). For assurers and investors, composition-based diagnostics can differentiate reports that are substantively informative from those that are selectively transparent. This distinction has economic relevance, as investors increasingly rely on sustainability information to assess risk exposure, resilience and long-term value creation (Pástor and Vorsatz 2020; Phang et al. 2023). For preparers, engaging candidly with sensitive topics—and documenting governance, targets and remediation—can enhance credibility and reduce future assurance frictions (Herzig 2006; Zdolšek and Beloglavec 2023).

Like all empirical studies, this research has limitations. The sample is restricted to U.S.-listed Forbes Global 2000 firms that consistently adopted the GRI framework. Although this population comprises very large and highly visible corporations, systematic GRI adopters are likely to be more engaged with formal SR. Accordingly, the findings may overstate the maturity of reporting practices and should be interpreted as reflecting the evolution of reporting strategies among GRI-committed large firms rather than the broader corporate population.

While firm fixed effects control for time-invariant characteristics, time-varying influences, such as regulatory developments, shifts in ESG-oriented investor pressure, and industry-specific shocks, may also shape reporting strategies. The results, therefore, provide descriptive evidence of evolving disclosure composition rather than causal effects. In addition, the analysis captures what is disclosed but not the depth, quantification or verifiability of disclosures. Future research could apply causal designs around regulatory milestones, incorporate time-varying firm controls, and integrate composition with measures of evidence quality, assurance and double materiality to further assess whether reporting is becoming substantively more informative.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

¹ This study reviewed all types of sustainability-related reports published by these companies under titles such as CSR report, ESG report, Citizenship report, etc.

² We justify the pre-COVID (2018–2019), acute COVID (2020–2021), and post-acute (2022–2023) periodisation based on World Health Organisation declarations and prior literature. We use the term post-acute to acknowledge that pandemic-related effects may persist beyond 2021; this period is not intended to imply a full return to pre-pandemic conditions, but rather a transition out of the acute emergency phase.

³ <https://www.ifrs.org/sustainability/knowledge-hub/introduction-to-issb-and-ifrs-sustainability-disclosure-standards/>.

⁴ <https://assets.kpmg.com/content/dam/kpmg/pl/pdf/2023/08/talkbook-get-ready-for-esrs.pdf>.

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